

Penetration Testing Report

Vstracam CCTV Camera Security Assessment

March 3, 2026

Report Date	March 3, 2026
Tester	[Your Name]
Target System	Vstracam IP Camera [Model/IP Address]
Assessment Type	Vulnerability Assessment
Status	Complete

1. Executive Summary

A security assessment was conducted on a Vstracam CCTV camera to identify potential vulnerabilities. The assessment discovered unauthorized access to camera control functions through exposed CGI commands, allowing remote manipulation of camera operation without authentication.

2. Scope and Methodology

2.1 Scope

Target: Vstracam IP Camera

Assessment Focus:

- HTTP/CGI endpoint security
- Authentication mechanisms
- Camera control functionality

2.2 Methodology

- Manual HTTP request analysis
- CGI parameter enumeration
- Authentication bypass testing

3. Findings

3.1 Finding 1: Unauthenticated Camera Control Access

Severity	High
Vulnerability Type	Broken Authentication / IDOR
CVSS Score	8.2 (High)
Affected Component	CGI control interface
Status	Confirmed
Date Discovered	March 3, 2026

3.1.1 Description

The Vstracam CCTV camera exposes CGI endpoints that allow remote users to control camera functions (pan, tilt, zoom) without requiring authentication. These endpoints can be accessed directly via HTTP requests with specific CGI parameters.

3.1.2 Technical Details

Vulnerable Endpoint:

- `/cgi-bin/api.cgi`

Example Vulnerable Request:

- `GET /cgi-bin/api.cgi?action=command&command=pan&value=1`

Parameters Discovered:

- `action=command`: Specifies the command to execute
- `command=pan|tilt|zoom`: Camera movement control
- `value=[int]`: Direction or speed parameter

3.1.3 Proof of Concept

Successfully executed:

1. Identified the CGI endpoint through HTTP traffic analysis.
2. Sent unauthenticated request to control pan/tilt functions.
3. Confirmed camera responded to movement commands.

3.1.4 Business Impact

An attacker could remotely control the camera without credentials, potentially redirecting surveillance or disabling monitoring. This poses a significant security risk in any environment relying on the camera for security purposes.

4. Remediation Recommendations

1. Implement authentication on all CGI endpoints (HTTP Basic Auth or token-based).
2. Require valid credentials for all camera control operations.
3. Implement rate limiting on control endpoints to prevent abuse.
4. Log all camera control requests with timestamp and source IP.
5. Consider using HTTPS instead of HTTP for encrypted communication.
6. Update firmware to the latest version if a patch is available.

5. Conclusion

The Vstracam CCTV camera contains a critical vulnerability allowing unauthenticated control of camera functions. Immediate remediation is recommended to secure the device and prevent unauthorized access.

6. Appendix: Additional Resources

6.1 CVSS v3.1 Vector

CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H

6.2 References

- OWASP Top 10 - A01:2021 Broken Access Control
- CWE-287: Improper Authentication
- CVSS Calculator: <https://www.first.org/cvss/calculator/3.1>